

Artificial intelligence-driven cardiac amyloidosis screening



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Transthyretin (ATTR) cardiac amyloidosis is an increasingly recognised disease, especially among older patients with aortic stenosis and heart failure.¹ Although the prognosis of patients with ATTR-cardiac amyloidosis is poor without treatment,¹ early initiation of targeted therapy reduces all-cause mortality and cardiovascular-related hospitalisations.² Bone scintigraphy imaging has emerged as a highly accurate method for identifying patients with ATTR-cardiac amyloidosis. However, early identification of patients with ATTR-cardiac amyloidosis remains a clinical challenge, with a median time from symptom onset to diagnosis of 2 years.³ Methods to automate opportunistic screening for ATTR-cardiac amyloidosis could potentially address this important care gap.

In *The Lancet Digital Health* Clemens P Spielvogel and colleagues⁴ present a generalisable solution for large-scale, opportunistic screening of bone scintigraphy imaging. The authors developed a multistep approach to identifying patients with abnormal uptake on planar imaging. First, image counts were normalised then deep learning was used to crop images of the thorax. The cropped images were used as the input for a convolutional neural network (CNN) tasked with classifying images as positive (abnormal planar uptake) or negative (no abnormal uptake). The overall predictive performance of the model was excellent (area under the curve [AUC] 1.000 [95% CI 1.000–1.000] during internal cross-validation) and remained relatively consistent across cohorts and radiotracers (AUC ranging from 0.925 to 1.000). The authors went on to evaluate common sources of false positive and negative results to further refine their approach, improving the positive predictive value from 0.886 to 0.932.

Although Spielvogel and colleagues' approach showed excellent performance, it is important to compare their study to a similar attempt published in 2023. Delbarre and colleagues trained a CNN to identify abnormal ^{99m}Tc-3,3'-diphosphono-1,2-propanodicarboxylic acid uptake on planar imaging, with AUC of 0.999 during internal cross-validation.⁵ The incremental value of the study by Spielvogel and colleagues is not only the larger population (16 241 vs 3048 patients), but also external testing of their approach in multiple centres with a variety of imaging protocols. External testing

is a critical step for any model,^{6,7} because it evaluates performance when deployed in new populations. With bone scintigraphy imaging, there are differences in image sets between sites related to radiotracers and imaging protocols, as highlighted by the drop in performance in sites using ^{99m}Tc-pyrophosphate. Additionally, it is worth noting that the ground truth used by Spielvogel and colleagues was established using a rigorous core laboratory approach, which involved each study being reviewed in triplicate (with another two readers for discrepancies). Although this core laboratory interpretation reflects a monumental amount of work, it is an important step given the high rate of discordance between physicians demonstrated in the multicase, multireader sample. Lastly, the authors finding that patients identified as having abnormal cardiac uptake were more likely to die during follow-up provides further confidence in the overall results given this uncertainty in the ground truth.

Although the test parameters of ^{99m}Tc-scintigraphy in the assessment of ATTR-cardiac amyloidosis might be excellent, there are a few important considerations regarding its use. The most important consideration is the known limitations of planar imaging for identifying patients with ATTR-cardiac amyloidosis.^{8,9} This limitation is particularly important in low prevalence populations, in which sensitivity and specificity of planar imaging is reduced.¹⁰ Therefore, this approach would be most effective if it was integrated into imaging workflows to automate the decision to acquire single-photon emission-CT imaging. This integrated method would allow more accurate screening, without requiring additional radiotracer injections. The second consideration is that methylene diphosphonate (MDP) is frequently used for general purpose bone scintigraphy imaging but is known to have reduced accuracy for ATTR-cardiac amyloidosis.¹ This reduced accuracy would limit the yield of screening in sites using MDP to identification of only patients with more advanced disease.

In summary, the authors developed and rigorously evaluated a model for the detection of abnormal radiotracer uptake on bone scintigraphy imaging. The generalisability demonstrated by the authors suggests that this model could be implemented clinically to assist in detecting patients with ATTR-cardiac amyloidosis.

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